Propositions

accompanying the dissertation

Structural reliability updating through proof load testing

by

Rein de Vries

1. Contrary to popular belief, proof load testing updates the distributions of all autocorrelated variables, not just the resistance.
   This proposition pertains to this dissertation.
2. When employed within a Bayesian reliability updating context, proof load testing provides uniquely accurate assessments.
   This proposition pertains to this dissertation.
3. It is always possible to assess structural reliability, even without information. In this case, the reliability index is zero, i.e. 50% chance of failure.
4. In a reliability-updating context, model uncertainties deserve primary attention, as they are typically poorly known yet highly influential.
5. New structures built largely from reused elements should not be required to adhere to the current reliability requirements for new structures.
6. Civil and structural engineers who dismiss statistics and computer science will soon become obsolete.
7. Aspiring PhD candidates should be encouraged to first obtain working experience; it maximises the study's value and helps them steer it.
8. Modern dissertations, predominantly linking open-access articles, are meant to be found and read online—printing them serves little purpose and wastes resources.
9. Vintage steel bikes keep improving their reliability: proof by cobbles. With carbon fibre, a single crack updates the frame to landfill.
10. Acting on prior knowledge is key to good outcomes. In finance, it can also lead to legal prosecution.

These propositions are regarded as opposable and defendable, and have been approved as such by the promotors prof. dr. ir. M. A. N. Hendriks, prof. dr. ir. R. D. J. M. Steenbergen, and prof. dr. ir. E. O. L. Lantsoght.